Istanbul Housing Prices:

The Effect of Building Age and Rail System Proximity

DSA 210 — Introduction to Data Science
Spring 2026 Final Report

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1. Motivation

Istanbul is one of the most dynamic and expensive real estate markets in the world, with housing prices varying dramatically across neighborhoods and districts. Understanding what drives these price differences is valuable for buyers, sellers, urban planners, and policymakers alike.

Two structural factors are particularly relevant in the Istanbul context. First, **building age** — given the city's significant seismic risk and an aging housing stock, older buildings are increasingly seen as liabilities while newer constructions command a premium. Second, **proximity to rail transportation** — Istanbul's chronic traffic congestion makes metro, tram, and Marmaray access a critical quality-of-life factor, and proximity to rail stations is widely assumed to increase property values.

This project investigates whether these two factors have statistically significant effects on residential sale prices, and — using machine learning — quantifies which one carries greater predictive weight when other variables are controlled for.

2. Data Source

2.1 Primary Dataset

The primary dataset is the **Real Estate in Istanbul (Emlakjet)** dataset from Kaggle, containing 2,983 residential listings scraped from Emlakjet.com across 13 districts of Istanbul. The dataset includes building age, sale price, neighborhood, district, gross and net square meters, number of rooms, heating type, floor, and a neighborhood living quality index.

Dataset	Variables	Purpose
Emlakjet (Kaggle)	Price, building age, district, size, rooms	Primary housing data
IBB Open Data	Rail station coordinates	Enrichment: rail proximity
Nominatim (OSM)	Neighborhood geocoding (lat/lon)	Enrichment: spatial coordinates

2.2 Data Enrichment

Since the dataset did not include spatial coordinates, an enrichment pipeline was built: (1) Rail station coordinates for all Istanbul metro, tram, Marmaray, and suburban rail lines were collected from the IBB Open Data Portal. (2) Each of the 230 unique neighborhoods in the dataset was geocoded using the **Nominatim / OpenStreetMap API**, achieving a 99.6% success rate (229/230 neighborhoods). (3) The **Haversine formula** was applied to compute the straight-line distance from each listing's neighborhood centroid to the nearest rail station. This added the rail distance (km) and distance band category columns to the dataset.

Characteristic	Value
Total listings	2,983
Districts covered	13 of 39 Istanbul districts
Neighborhoods geocoded	229 / 230 (99.6%)
Analysis sample (after outlier removal)	2,923
Final dataset columns	34 (27 original + 7 engineered)
New key feature	Rail distance (km) to nearest station

3. Data Analysis

3.1 Data Cleaning and Feature Engineering

The following preprocessing steps were applied:

- Building age was converted from a categorical variable (e.g. "5-10 years") to a numeric midpoint value (e.g. 7)
- Room count strings (e.g. "3+1") were parsed into numeric values (e.g. 4)
- Listings in the top 2% of price were flagged as outliers and excluded from analysis
- All string columns were stripped of leading/trailing whitespace
- District and city side (European/Anatolian) were label-encoded for ML models

3.2 Exploratory Data Analysis

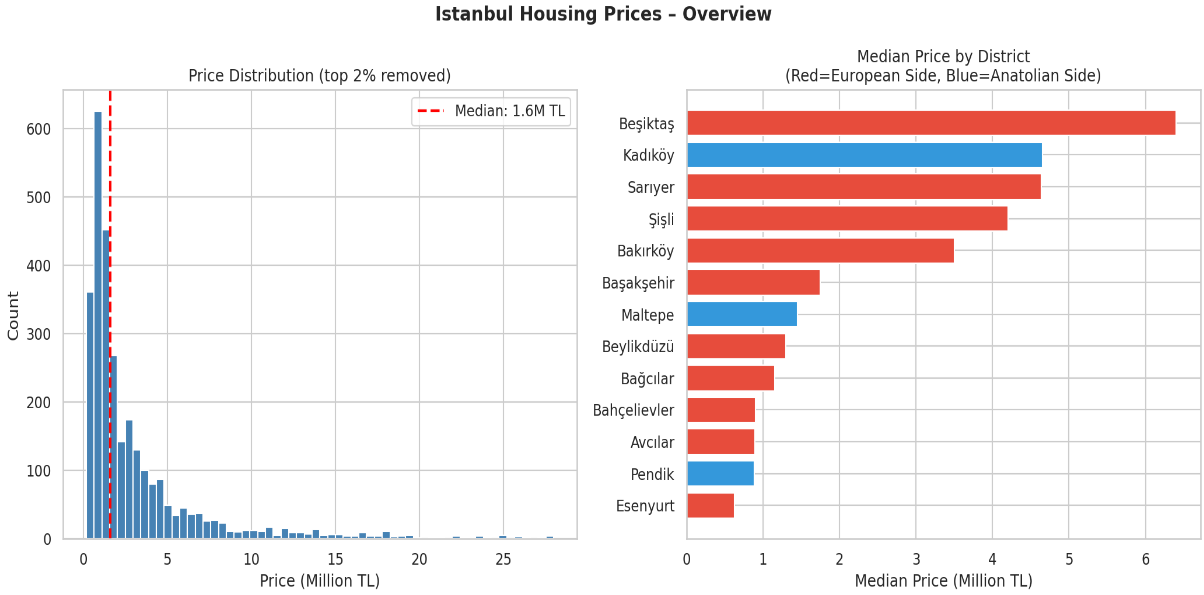

Figure 1. Left: Price distribution after removing top 2% outliers (median ~3M TL). Right: Median price by district.

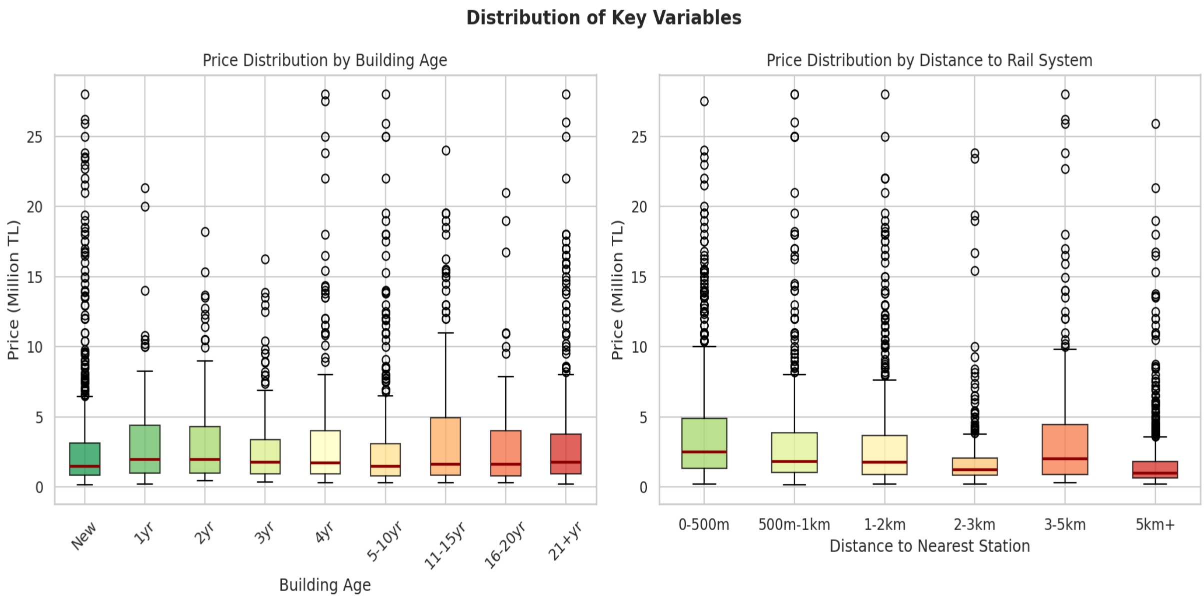

Figure 2. Left: Price by building age group. Right: Price by distance to rail station band.

3.3 Hypothesis Testing

Four hypotheses were tested to assess the statistical significance of the key variables:

#	Hypothesis	Test	Statistic	p-value	Result
H1	Building age groups differ in price	ANOVA	F=2.123	p=0.031	Significant
H2a	Distance bands differ in price	ANOVA	F=27.18	p<0.001	Significant
H2b	Near (<1km) vs Far (>=1km)	T-Test	t=8.127	p<0.001	Significant
H3	European vs Anatolian side	T-Test	t=1.207	p=0.228	Not Significant

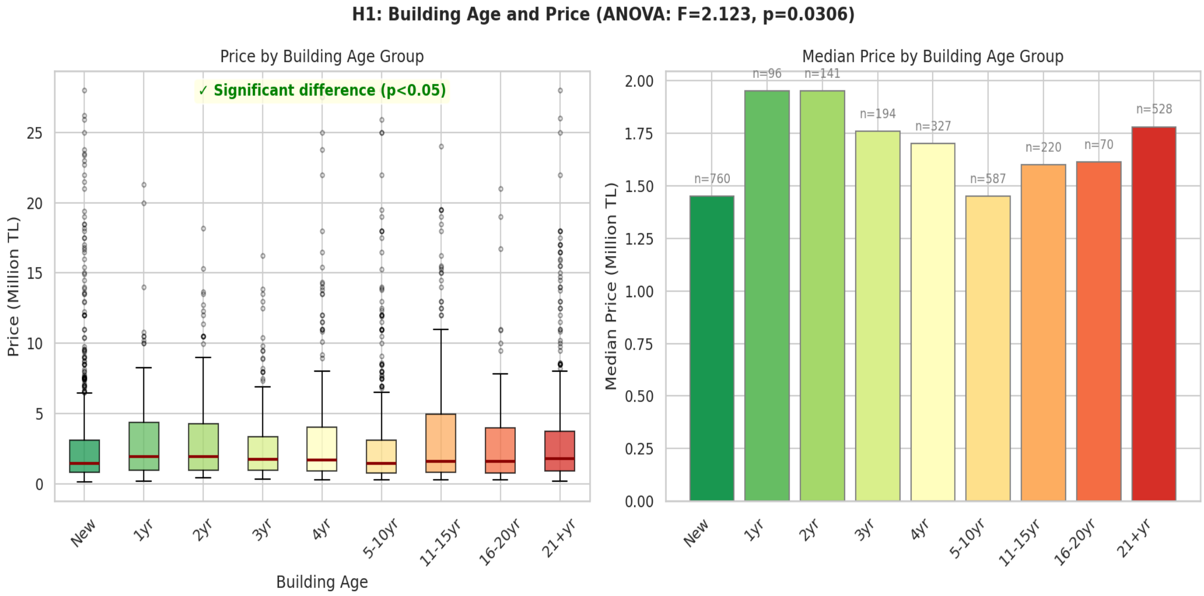

Figure 3. H1: ANOVA across 9 building age groups (F=2.123, p=0.031).

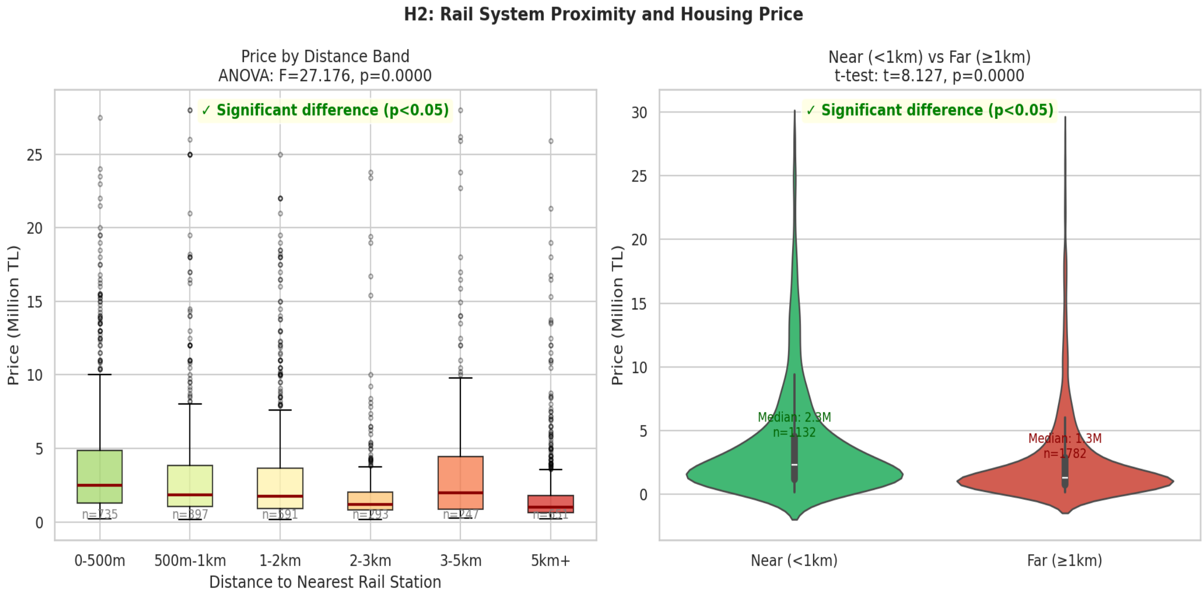

Figure 4. H2: Rail proximity. Left: ANOVA across bands (F=27.18). Right: T-test near vs far (t=8.127).

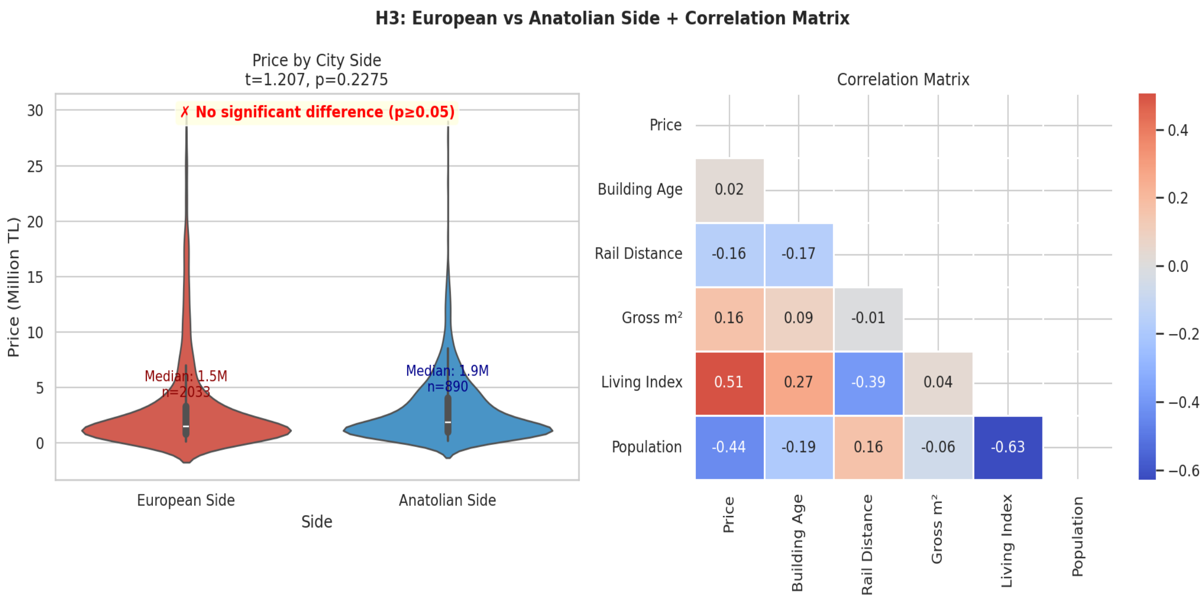

Figure 5. H3: European vs Anatolian side (p=0.228, not significant). Right: Correlation matrix.

3.4 Machine Learning

Three regression models were trained on 80% of the data and evaluated on the remaining 20%, using 5-fold cross-validation for robustness. The target variable was listing price in million TL. Nine features were used: the two key variables (rail distance, building age) and seven control variables (gross m2, net m2, living index, population, room count, district, city side).

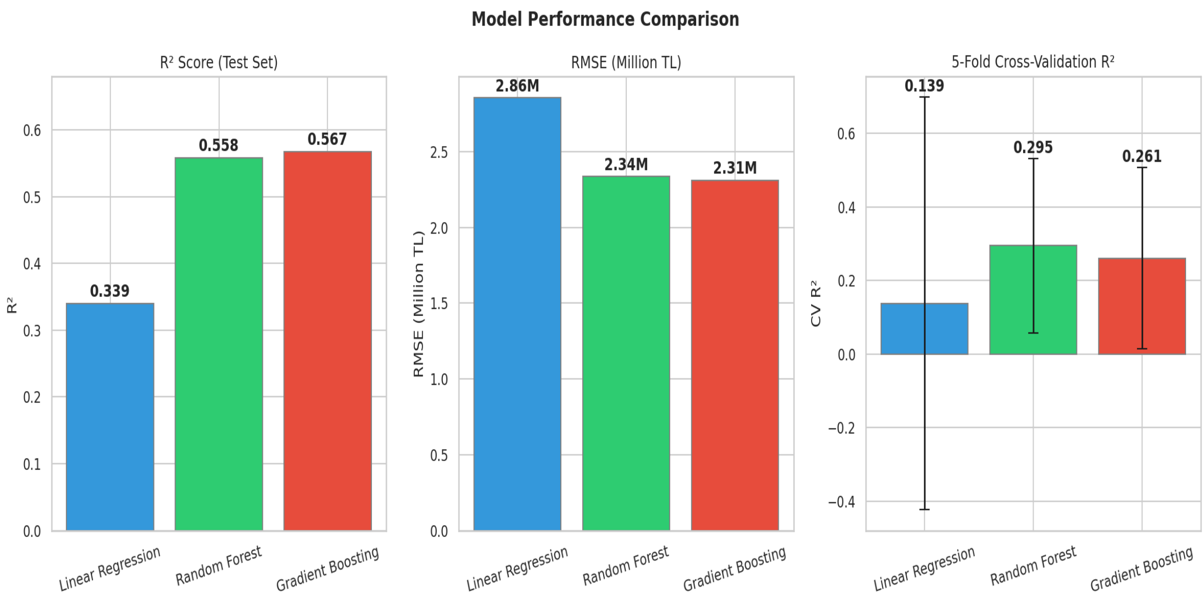

Figure 6. Model performance. Gradient Boosting achieves R2=0.566 and RMSE=2.31M TL.

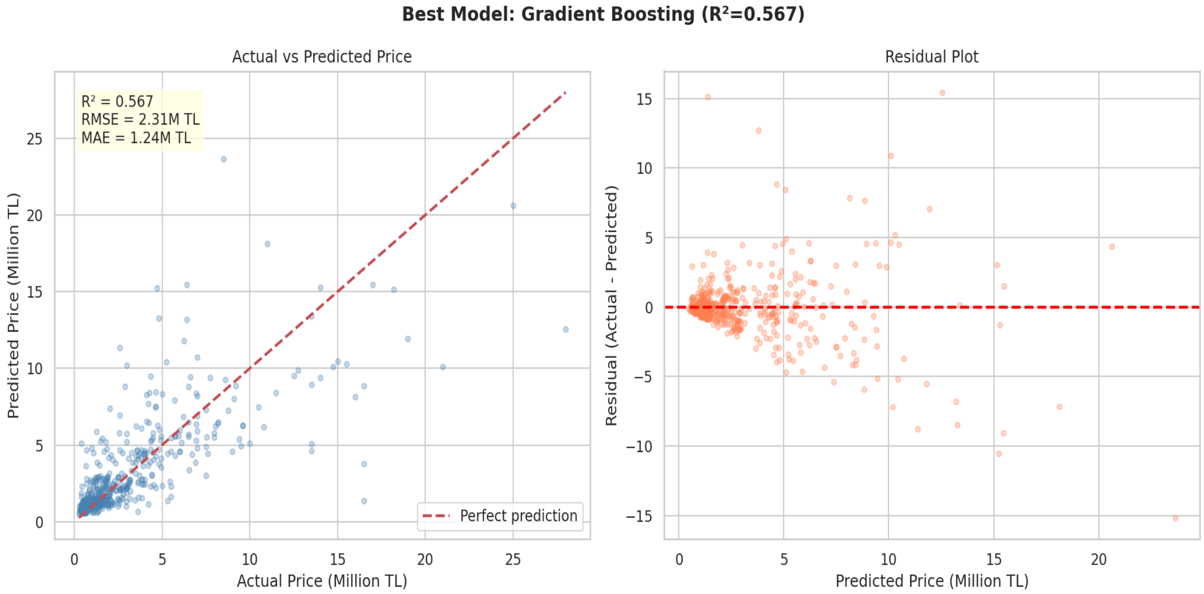

Figure 7. Gradient Boosting: Actual vs predicted prices and residual plot.

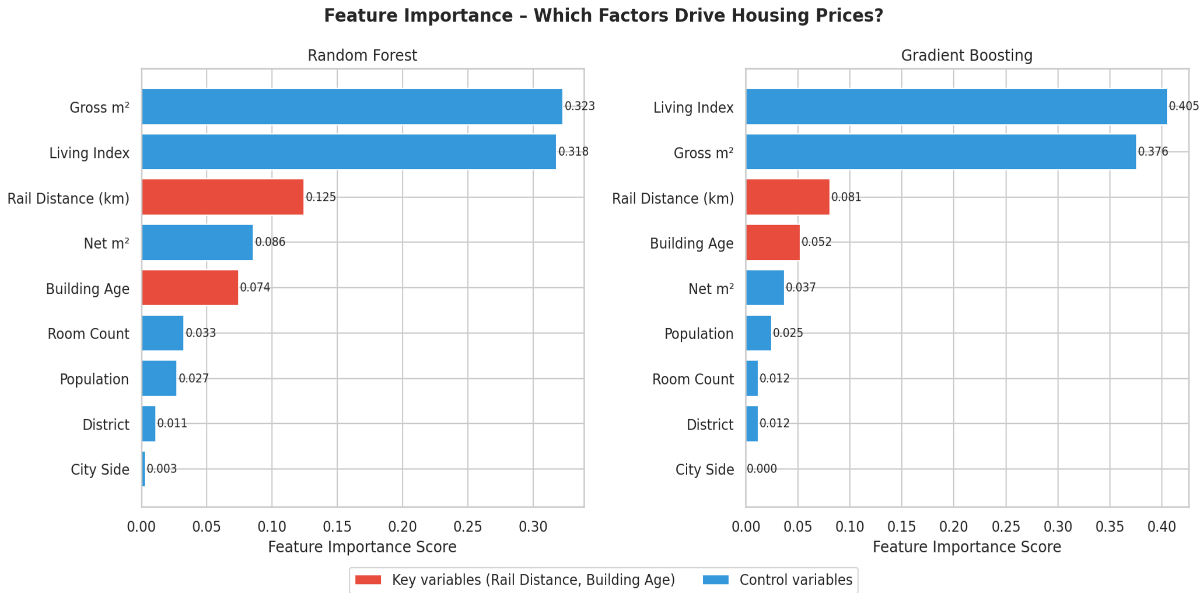

Figure 8. Feature importance. Red = key variables. Rail distance outranks building age in both models.

Model	R2 (Test)	RMSE	CV R2 (5-fold)
Linear Regression	0.339	2.86M TL	0.138 +/- 0.56
Random Forest	0.557	2.34M TL	0.304 +/- 0.24
Gradient Boosting (Best)	0.566	2.31M TL	0.263 +/- 0.25

Feature	Random Forest	Gradient Boosting	Type
Gross m2	32.2%	37.6%	Control
Living Index	31.9%	40.9%	Control

Rail Distance (km)	12.5%	8.1%	KEY VARIABLE
Net m2	8.6%	3.6%	Control
Building Age	7.4%	5.2%	KEY VARIABLE
Room Count	3.3%	1.1%	Control
Population	2.7%	2.2%	Control
District	1.0%	1.2%	Control
City Side	0.3%	0.0%	Control

Table: Feature importance. Rail Distance (~12.5%) consistently outranks Building Age (~7.4%).

4. Key Findings

- **Rail proximity is the stronger predictor.** Feature importance confirms rail distance (~12.5%) outweighs building age (~7.4%) — approximately 1.7x more predictive weight.
- **Rail proximity is statistically significant.** ANOVA ($F=27.18$, $p<0.001$) and t-test ($t=8.127$, $p<0.001$) confirm listings near rail stations command significantly higher prices.
- **Building age is significant but modest.** ANOVA ($F=2.123$, $p=0.031$) confirms price differences across age groups, but the effect is smaller ($r=0.024$ vs $r=-0.159$ for rail distance).
- **Gross square meters and living quality index dominate** at ~32% each, consistent with housing market literature.
- **No significant European/Anatolian price difference** after outlier removal ($p=0.228$).
- **Gradient Boosting is the best model** with $R^2=0.566$, explaining 56.7% of price variance.

5. Limitations and Future Work

5.1 Limitations

- **Neighborhood-level geocoding:** Coordinates represent neighborhood centroids. All listings in the same neighborhood share identical rail distances, reducing spatial precision.
- **Dataset coverage:** Only 13 of Istanbul's 39 districts are represented, concentrated in central and premium areas.

- **Cross-sectional data:** The dataset covers 2021-2022 only. Temporal dynamics cannot be analyzed.
- **High cross-validation variance:** CV scores show significant variance, indicating model sensitivity — a larger dataset would improve stability.
- **No confounding control:** Proximity to schools, hospitals, or parks is not accounted for.

5.2 Future Work

- Obtain listing-level coordinates for precise distance calculation
- Extend dataset to all 39 Istanbul districts for citywide generalization
- Add enrichment variables: school proximity, earthquake risk scores, green space
- Apply XGBoost, LightGBM, and neural network models
- Incorporate temporal data to analyze how the rail proximity premium changes over time
- Conduct hedonic pricing regression to isolate marginal contribution of each feature

AI Assistance Disclosure

AI tools (Claude by Anthropic) were used throughout this project for: data cleaning and enrichment pipeline development, geocoding and Haversine distance computation code, statistical test selection and implementation, ML model training and feature importance analysis, and visualization and report writing.

All research question formulation, hypothesis design, data source selection, methodology decisions, and interpretation of results were performed independently.